

Crop Tracking System (Crop-tracker) Specification

This document serves as the primary Software Requirements Specification (SRS) and Architectural Blueprint for the **Crop Tracking Project**. It outlines system objectives, core features, database schema definitions, technical architecture, and implementation milestones.

1. Executive Summary & Project Vision

The Crop Tracking System (*Crop-tracker*) is an integrated agricultural management application designed to help farmers, agronomists, and farm managers monitor crop growth cycles, manage soil health, track field activities (planting, fertilization, pest control, harvesting), and analyze yield performance over time.

2. System Requirements & Feature Matrix

The table below provides a summary of core functional requirements, user permissions, and priority levels across system modules:

Module	Feature Description	Target Roles	Priority
Farm & Field Setup	Map farm boundaries, define plots/zones, record soil composition and irrigation types.	Farm Admin, Owner	High
Crop Cycle Management	Track growth stages from seed planting to harvesting, log expected vs	Agronomist, Worker	High

Module	Feature Description	Target Roles	Priority
	actual yields.		
Activity Logging	Log fertilizer applications, pesticide treatments, watering schedules, and labor tasks.	Field Worker	Medium
Analytics & Reporting	Generate yield reports, historical trend comparison, input cost analysis, and harvest forecasting.	Manager, Owner	Medium

3. Database Schema & Data Models

Below is the relational data model representation for core entities in JSON schema format:

```
{
  "Field": {
    "field_id": "UUID",
    "name": "String",
    "area_hectares": "Float",
    "soil_type": "String",
    "location": { "lat": "Float", "lng": "Float" }
  },
  "CropCycle": {
    "cycle_id": "UUID",
    "field_id": "UUID",
```